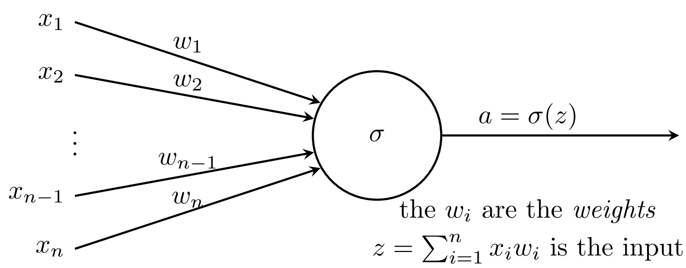
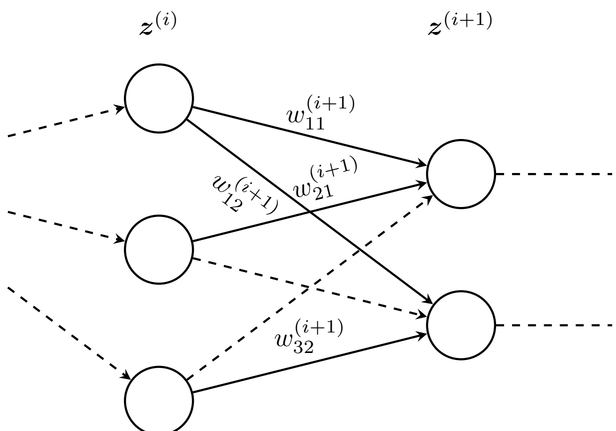


the w_i are the *weights*

$z = \sum_{i=1}^{n} x_i w_i$ is the input

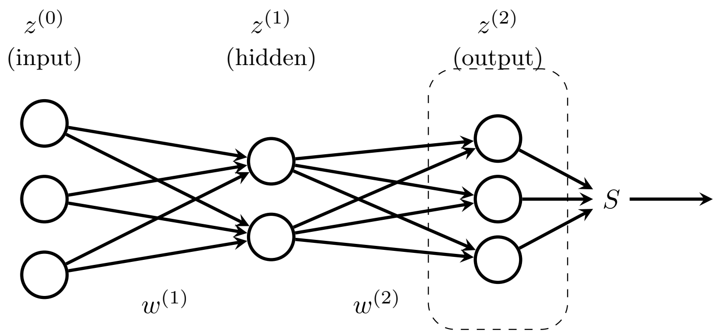
$a = \sigma(z)$ is the activation or output

$$\sigma(x) = \frac{1}{1+e^{-x}} \text{ or } \sigma(t) = \tanh(x) \text{ or } \sigma(t) = \begin{cases} t & t \geq 0 \\ 0 & t < 0 \end{cases} \text{ or others}$$

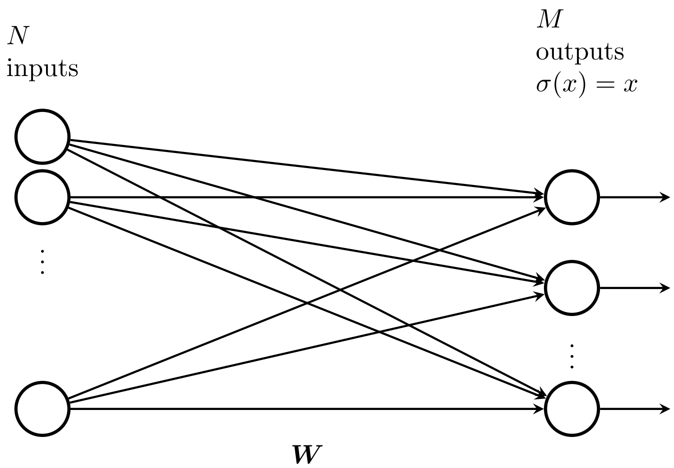


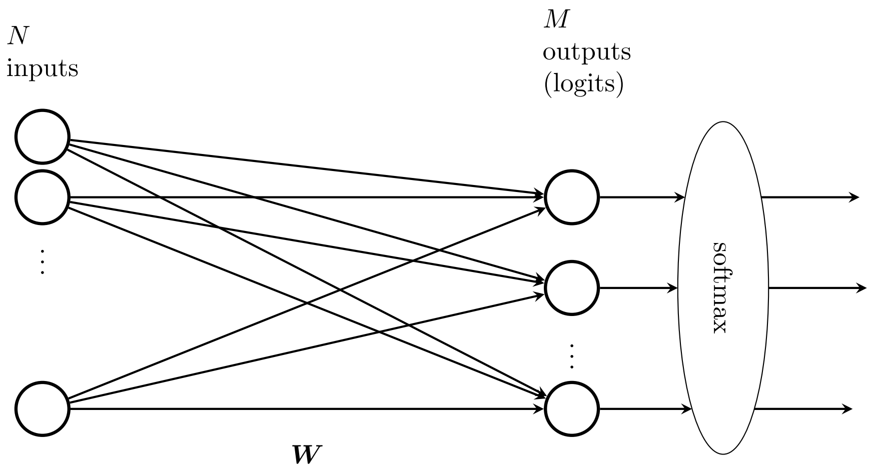
$$z_\ell^{(i+1)} = \sum_{j=1}^{n^{(i)}} a_j^{(i)}\, w_{j\ell}^{(i+1)}$$

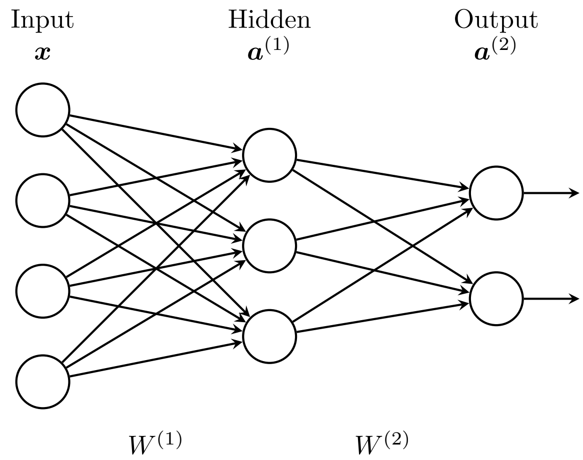
$$a_\ell^{(i+1)} = \sigma\big(z_\ell^{(i+1)}\big)$$

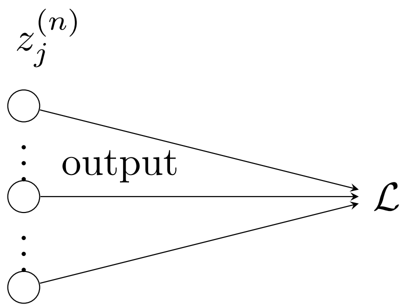


$z^{(0)}$	1×3 matrix
$w^{(1)}$	3×2 matrix
$w^{(2)}$	2×3 matrix


N
inputs
M
outputs
$\sigma(x) = x$
$\boldsymbol{W}$


N
inputs
M
outputs
(logits)
softmax
$\boldsymbol{W}$


Input
$\boldsymbol{x}$
Hidden
$\boldsymbol{a}^{(1)}$
Output
$\boldsymbol{a}^{(2)}$
$W^{(1)}$
$W^{(2)}$



$$\delta_j^{(n)} = \frac{\partial \mathcal{L}}{\partial z_j^{(n)}}$$